Grade 6 Accelerated
Unit 1
Lesson 1 Homework

Name Answer Key__________
Date______________________
Period_____________________

Find the greatest common factor (GCF) of each pair of numbers.

1. 20 and 32

$20: 1, 2, 4, 5, 10, 20$
$32: 1, 2, 4, 8, 16, 32$

GCF $= 4$

2. 60 and 486

$60: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60$
$486: 1, 2, 3\ 6, 9, 18, 27, 54, 81, 162, 243, 486$

GCF $= 6$

3. 13 and 9

$13: 1, 13$
$9: 1, 3, 9$

GCF $= 1$

4. 392 and 616

$392: 1, 2, 4, 7, 8, 14, 28, 49, 56, 98, 196, 392$
$616: 1, 2, 4, 7, 8, 11, 14, 22, 28, 44, 56, 77, 88, 154,$
$308, 616$

GCF $= 56$

5. Dr. Harris has 90 sixth-grade students and 75 seventh-grade students in her chorus class. She wants to make groups of performers, with the same amount of sixth- and seventh-grade students in each group; keeping in mind that she wants to form as many groups as possible. What is the largest number of groups that could be formed?

$90: 1, 2, 3, 5, 6, 9, 10, 15, 18, 30, 45, 90$
$75: 1, 3, 5, 15, 25, 75$

The largest group can have 15 students.

Rewrite each fraction in simplest form using the GCF of the numerator and denominator.

6. $\frac{30}{55}$

$\frac{30}{55} = \frac{\cancel{5} \times 6}{\cancel{5} \times 11} = \frac{6}{11}$

7. $\frac{17}{68}$

$\frac{17}{68} = \frac{1 \times \cancel{17}}{4 \times \cancel{17}} = \frac{1}{4}$

8. $\frac{306}{630}$

$\frac{306}{630} = \frac{\cancel{18} \times 17}{\cancel{18} \times 35} = \frac{17}{35}$

9. $\frac{72}{244}$

$\frac{72}{244} = \frac{18 \times \cancel{4}}{61 \times \cancel{4}} = \frac{18}{61}$

10. Marguerite wanted to rewrite the fraction $\frac{760}{800}$ in simplest form. She said, "The GCF of 760 and 800 is 10. Since $760 \div 10 = 76$ and $800 \div 10 = 80$, I can rewrite $\frac{760}{800}$ in simplest form as $\frac{76}{80}$." Write a note to Marguerite explaining why she is incorrect and what she can do to correct her mistake.

It can be simplified even more.

$$\frac{76}{80} = \frac{19 \times \cancel{4}}{20 \times \cancel{4}} = \frac{19}{20}$$

Grade 6 Accelerated
Unit 1
Lesson 2 Homework

Name Answer Key________________
Date_______________________
Period______________________

Find the least common multiple (LCM) of each pair of numbers.

1. 13 and 3

$3: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39$
$13: 13, 26, 39$

LCM $= 39$

2. 18 and 12

$12: 12, 24, 36$
$18: 18, 36$

LCM $= 36$

3. 4 and 20

$4: 4, 8, 12, 16, 20$
$20: 20$

LCM $= 20$

4. 8 and 25

$8: \ldots 160, 168, 176, 184, 192, 200$
$25: 25, 50, 75, 100, 125, 150, 175, 200$

LCM $= 200$

5. 15 and 20

$15: 15, 30, 45, 60$
$20: 20, 40, 60$

LCM $= 60$

6. 10 and 14

$10: 10, 20, 30, 40, 50, 60, 70$
$14: 14, 28, 42, 56, 70$

LCM $= 70$

7. On a machine, a green light blinks every 4 seconds and a yellow light blinks every 5 seconds. If the two lights just blinked at exactly the same time, after how many seconds will both lights blink at the same time again?

$4: 4, 8, 12, 16, 20$
$5: 5, 10, 15, 20$

After 20 seconds both lights will blink at the same time again.

8. A principal is making welcome packets for his students with one pencil and one eraser in each. The pencils come in packs of 24 and the erasers come in packs of 18. The principal bought packs of pencils and packs of erasers such that he had the same number of pencils and erasers to make welcome packets. What is the smallest number of welcome packets the principal could have made?

$18: 18, 36, 54, 72$
$24: 24, 48, 72$

The smallest number of welcome packets is 72.

9. An electronics store is doing a promotion where every 20^{th} customer gets a $\$10$ gift card and every 25^{th} customer gets a pair of Bluetooth ear buds. Which number customer would be the first to receive both the gift card and the ear buds?

$20: 20, 40, 60, 80, 100$
$25: 25, 50, 75, 100$

The 100^{th} customer will receive both.

10. Heiran and Jose both work long shifts in a hospital. Heiran has a day off every 3 days and Jose has a day off every 5 days. If they are both off today, after how many days will they both have the same day off again?

$3: 3, 6, 9, 12, 15$
$5: 5, 10, 15$

They will both have off again in 15 days.

Grade 6 Accelerated
Unit 1
Lesson 3 Homework

Name______________________
Date______________________
Period______________________

Complete the table by rewriting each multiplication expression using the distributive property, modeling with area, and finding the product.

Multiplication Expression	Distributive Property	Area Model	Product
1. 37×9	$9(30+7)$ $= 270 + 63$ $= 333$	__30__ + __7__ __9__ | 270 | 63	333
2. 8×54	$8(50+4)$ $= 400 + 32$ $= 432$	__50__ + __4__ __8__ | 400 | 32	432
3. 95×6	$6(90+5)$ $= 540 + 30$ $= 570$	__90__ + __5__ __6__ | 540 | 30	570

For each expression, generate an equivalent numeric expression using the distributive property and then find the value.

4. 7×23
$7(20+3)$
$= 7 \times 20 + 7 \times 3$
$= 140 + 21$
$= 161$

5. $15 \cdot 9$
$9(10+5)$
$= 9 \times 10 + 9 \times 5$
$= 90 + 45$
$= 135$

6. 5×48
$5(40+8)$
$= 5 \times 40 + 5 \times 8$
$= 200 + 40$
$= 240$

Rewrite an equivalent expression for each sum using the distributive property with the greatest common factor (GCF).

7. $63 + 105$

$21(3 + 5)$

8. $270 + 414$

$18(15 + 23)$

9. Generate two different equivalent expressions for the sum $168 + 216$ by rewriting it using the distributive property with two different common factors.

Answers may vary.

$12(14 + 18)$
$24(7 + 9)$

10. Zoriah was using the distributive property to write an expression equivalent to $108 + 16$. Her expression was $4(27 + 16)$. Is Zoriah correct? If yes, explain how she determined her expression. If not, explain Zoriah's error and write a correct equivalent expression.

Her expression is incorrect. $4 \times 27 = 108$ but $4 \times 16 = 64$ not 16. If she wanted to use the common factor of 4, the correct expression would be $4(27 + 4)$.

Grade 6 Accelerated
Unit 1
Lesson 4 Homework

Name Answer Key
Date
Period

Find each product using the area model.

1. $(74.3)(9.2) = 683.56$

	9	0.2
74	666	14.8
0.3	2.7	0.06

2. $(2.05)(30.8) = 63.14$

	30	0.8
2	60	1.6
0.05	1.5	0.04

3. Find the product 27.9×8.3 using partial products.

$$
\begin{array}{r}
27.9 \\
\times\ 8.3 \\
\hline
0.27 \\
2.1 \\
6.0
\end{array}
\left.\vphantom{\begin{array}{r}0\\0\\0\end{array}}\right] = 8.37
$$

$$
\begin{array}{r}
7.2 \\
56.0 \\
160.0 \\
\hline
231.57
\end{array}
\left.\vphantom{\begin{array}{r}0\\0\\0\end{array}}\right] = 223.2
$$

Find the product $61.5 \cdot 3.84$ using the specified method.

<table>
<tr><th>4. Partial Products</th><th>5. Standard Algorithm</th></tr>
<tr><td>$$\begin{array}{r} 61.5 \\ \times\ 3.84 \\ \hline 0.02 \\ 0.04 \\ 2.4 \end{array} \Bigg] = 2.46$$
$$\begin{array}{r} 0.4 \\ 0.8 \\ 48.0 \end{array} \Bigg] = 49.2$$
$$\begin{array}{r} 1.5 \\ 3.0 \\ 180.0 \end{array} \Bigg] = 184.5$$
$$236.16$$</td><td>$$\begin{array}{r} 61.5 \\ \times\ 3.84 \\ \hline 2.46 \\ 49.2 \\ 184.5 \\ \hline 236.16 \end{array}$$</td></tr>
</table>

Find each product using any method.

6. $(5.93)(8.1) = 48.033$

	8	0.1
5	40	0.5
0.93	7.44	0.093

7. $183.5 \times 7.28 = 1{,}335.88$

$$\begin{array}{r} 183.5 \\ \times\ 7.28 \\ \hline 14.68 \\ 36.7 \\ 1284.5 \\ \hline 1335.88 \end{array}$$

8. $1.523 \cdot 4.9 = 7.4627$

	4	0.9
1	4	0.9
0.523	2.092	0.4707

9. $(41.9)(0.0353) = 1.47907$

$$\begin{array}{r} 41.9 \\ \times\ 0.0353 \\ \hline 0.01257 \\ 0.2095 \\ 1.257 \\ \hline 1.47907 \end{array}$$

10. Giordana made a mistake when using the standard algorithm to find the product 29.64×0.15. Write Giordana a note explaining her mistake.

$$\begin{array}{r} \mathbf{29.64} \\ \times\ \mathbf{0.15} \\ \hline 148.20 \\ +\ 29.64 \\ \hline \mathbf{177.84} \end{array}$$

The decimals are both misplaced. The first should be 1.482, the second is 2.964, all of which would give a final product of 4.446.

Grade 6 Accelerated
Unit 1
Lesson 5 Homework

Name Answer Key__________
Date______________________
Period_____________________

Find each quotient. Show your work.

1. $238.6 \div 0.4 = 596.5$

$$\begin{array}{r} 596.5 \\ 04\overline{)2386.0} \\ \underline{-20} \\ 38 \\ \underline{-36} \\ 26 \\ \underline{-24} \\ 20 \\ \underline{-20} \\ 0 \end{array}$$

2. $61.36 \div 5.2 = 11.8$

$$\begin{array}{r} 11.8 \\ 52\overline{)613.6} \\ \underline{-52} \\ 93 \\ \underline{-52} \\ 416 \\ \underline{-416} \\ 0 \end{array}$$

3. $106.33 \div 0.49 = 217$

$$\begin{array}{r} 217 \\ 049\overline{)10633} \\ \underline{-98} \\ 63 \\ \underline{-49} \\ 143 \\ \underline{-143} \\ 0 \end{array}$$

4. $24.418 \div 4.21 = 5.8$

$$\begin{array}{r} 5.8 \\ 421\overline{)2441.8} \\ \underline{-2105} \\ 3368 \\ \underline{-3368} \\ 0 \end{array}$$

5. $184.5 \div 10.25 = 18$

$$\begin{array}{r} 18 \\ 1025\overline{)18450} \\ \underline{-1025} \\ 8200 \\ \underline{-8200} \\ 0 \end{array}$$

6. $27.531 \div 43.7 = 0.63$

$$\begin{array}{r} 0.63 \\ 437\overline{)275.31} \\ \underline{-2622} \\ 1311 \\ \underline{-1311} \\ 0 \end{array}$$

7. Henrietta was finding the quotient $178.2 \div 3.24$, but she knows her answer doesn't make sense. Her work is shown. Write a note to Henrietta explaining her mistake.

$$
\begin{array}{r}
5.5 \\
324\,\overline{)\,1782} \\
-1620 \\
\hline
162\ \ 0 \\
-162\ \ 0 \\
\hline
000\ \ 0
\end{array}
$$

The divisor is multiplied by 100 to get 324, the dividend also needs to be multiplied by 100 to get 17820. The quotient should be 55.

An expression is shown. Use the expression to answer questions 8-10.

$$120.9 \div 0.62$$

8. Which of the expression is equivalent to $120.9 \div 0.62$? Select all that apply.

- ☐ $1209 \div 62$
- ☒ $1209 \div 6.2$
- ☐ $12.09 \div 6.2$
- ☒ $12090 \div 62$
- ☐ $12090 \div 620$

9. Select one of your answer choices from question 8 and explain why it is equivalent to $120.9 \div 0.62$.

Answers may vary.

$12090 \div 62$ is equivalent to $120.9 \div 0.62$ because both the dividend and divisor have been multiplied by 100.

10. Find the quotient $120.9 \div 0.62$.

$$
\begin{array}{r}
195 \\
062\,\overline{)\,12090} \\
\underline{-62} \\
589 \\
\underline{-558} \\
310 \\
\underline{-310} \\
0
\end{array}
$$

Grade 6 Accelerated
Unit 1
Lesson 6 Homework

Name Answer Key__________
Date_______________________
Period______________________

Find each product using the area model.

1. $2\frac{2}{3} \times 3\frac{1}{5} = 8\frac{8}{15}$

	2	$\frac{2}{3}$
3	6	2
$\frac{1}{5}$	$\frac{2}{5} = \frac{6}{15}$	$\frac{2}{15}$

2. $6\frac{1}{2} \times 2\frac{1}{4} = 13\frac{13}{8} = 14\frac{5}{8}$

	6	$\frac{1}{2}$
2	12	1
$\frac{1}{4}$	$\frac{6}{4} = \frac{12}{8}$	$\frac{1}{8}$

3. $4\frac{3}{4} \times 8\frac{1}{2} = 40\frac{3}{8}$

	4	$\frac{3}{4}$
8	32	6
$\frac{1}{2}$	2	$\frac{3}{8}$

Find each product. Rewrite each product in simplest form. Show your work.

4. $\frac{7}{20} \times \frac{2}{14} = \frac{1}{20}$

$= \frac{14}{280} = \frac{1 \times \cancel{14}}{20 \times \cancel{14}} = \frac{1}{20}$

5. $\frac{5}{12} \times 4\frac{1}{2} = 1\frac{7}{8}$

$= \frac{5}{12} \times \frac{9}{2} = \frac{45}{24} = \frac{15 \times \cancel{3}}{8 \times \cancel{3}} = \frac{15}{8} = 1\frac{7}{8}$

6. $3\frac{5}{8} \times 2\frac{3}{5} = 9\frac{17}{40}$

$= \frac{29}{8} \times \frac{13}{5} = \frac{377}{40} = 9\frac{17}{40}$

7. $\frac{21}{100} \times 2\frac{2}{3} = \frac{14}{25}$

$= \frac{21}{100} \times \frac{8}{3} = \frac{168}{300} = \frac{14 \times \cancel{12}}{25 \times \cancel{12}} = \frac{14}{25}$

8. $8\frac{7}{15} \times \frac{3}{4} = 6\frac{7}{20}$

$= \frac{127}{15} \times \frac{3}{4} = \frac{381}{60} = \frac{127 \times \cancel{3}}{20 \times \cancel{3}} = \frac{127}{20} = 6\frac{7}{20}$

9. $5\frac{18}{25} \times 3\frac{1}{2} = 20\frac{1}{50}$

$= \frac{143}{25} \times \frac{7}{2} = \frac{1001}{50} = 20\frac{1}{50}$

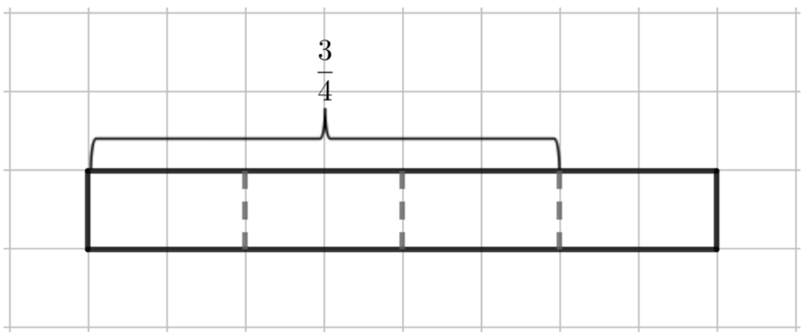
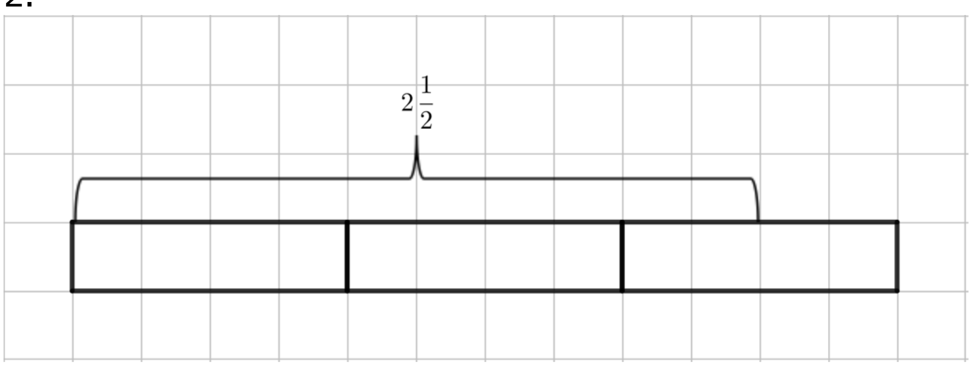
10. Arlo and Breanna were working on the multiplication expression $4\frac{5}{9} \times 3\frac{2}{5}$. Arlo says that the product is $12\frac{10}{45}$ or $12\frac{2}{9}$ because $4 \times 3 = 12$ and $\frac{5}{9} \times \frac{2}{5} = \frac{10}{45}$. Breanna says that the product is $\frac{697}{45}$ or $15\frac{22}{45}$ because $\frac{41}{9} \times \frac{17}{5} = \frac{697}{45}$. Explain who is correct and the mistake that the other student made.

Breanna is correct because you can't separate the whole numbers from the fractions when multiplying mixed numbers. You have to put them in improper fractions, like Breanna, before multiplying them.

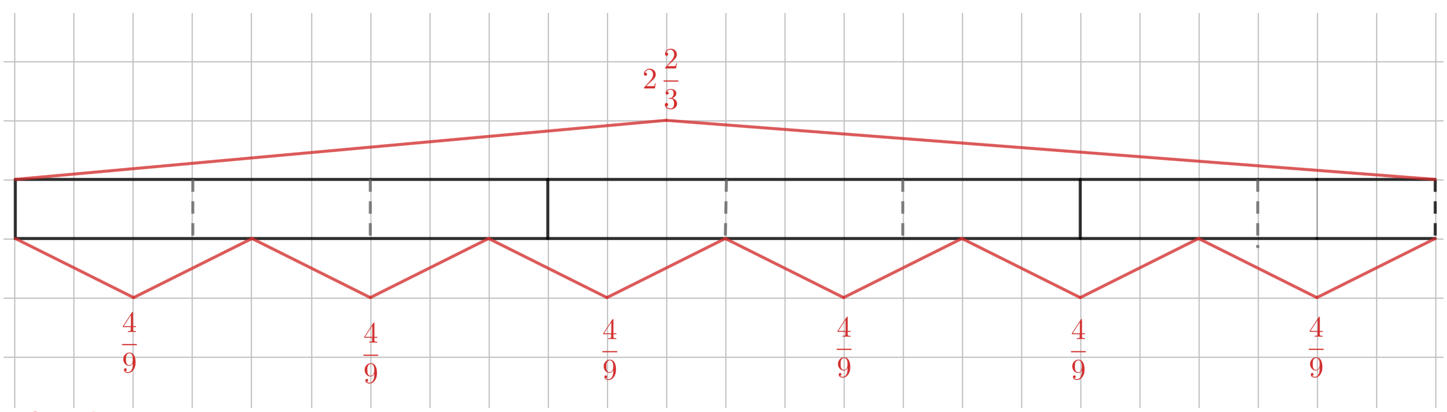
Grade 6 Accelerated
Unit 1
Lesson 7 Homework

Name Answer Key__________
Date________________________
Period______________________

Complete the table by finding each quotient. Use the tape diagram shown.

Tape Diagram	Division Expression	Quotient
1. 	$\frac{3}{4} \div \frac{3}{8}$	2
2. 	$2\frac{1}{2} \div \frac{3}{4}$	$3\frac{1}{3}$

3. Draw a tape diagram to find the quotient of $2\frac{2}{3} \div \frac{4}{9}$.



$2\frac{2}{3} \div \frac{4}{9} = 6$

Find each quotient. Simplify when possible.

4. $\frac{2}{5} \div \frac{6}{7} = \frac{7}{15}$

$= \frac{2}{5} \times \frac{7}{6} = \frac{14}{30} = \frac{7}{15}$

5. $\frac{6}{7} \div \frac{2}{5} = 2\frac{1}{7}$

$= \frac{6}{7} \times \frac{5}{2} = \frac{30}{14} = \frac{15}{7} = 2\frac{1}{7}$

6. $\frac{10}{13} \div 5 = \frac{2}{13}$

$= \frac{10}{13} \times \frac{1}{5} = \frac{10}{65} = \frac{2}{13}$

7. $8 \div \frac{4}{15} = 30$

$= \frac{8}{1} \times \frac{15}{4} = \frac{120}{4} = 30$

8. $6\frac{4}{5} \div \frac{9}{10} = 7\frac{5}{9}$

$= \frac{34}{5} \times \frac{10}{9} = \frac{340}{45} = \frac{68}{9} = 7\frac{5}{9}$

9. $4\frac{1}{8} \div 1\frac{5}{16} = 3\frac{1}{7}$

$= \frac{33}{8} \times \frac{16}{21} = \frac{528}{168} = \frac{22}{7} = 3\frac{1}{7}$

10. Elijah used the standard algorithm to divide $3\frac{5}{6} \div 2\frac{1}{2}$. His work is shown. Is Elijah correct? If he is correct, explain the steps that he took. If he is incorrect, explain his mistake and fix it.

Step 1: $3\frac{5}{6} \div 2\frac{1}{2}$

Step 2: $\frac{23}{6} \div \frac{5}{2}$

Step 3: $\frac{6}{23} \times \frac{5}{2}$

Step 4: $\frac{30}{46}$ or $\frac{15}{23}$

Elijah is incorrect because he used the reciprocal of the first fraction when he only should have used the reciprocal of the second fraction.
The correct answer is:
$\frac{23}{6} \times \frac{2}{5} = \frac{46}{30} = \frac{23}{15} = 1\frac{8}{15}$